

Qualitative Research in the Era of AI: A Return to Positivism or a New Paradigm?

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journals.sagepub.com/home/ijq**Georgios Chatzichristos¹**

Abstract

The integration of artificial intelligence (AI) into qualitative research is transforming the landscape of social inquiry, raising significant epistemological and methodological questions. This study explores the dual potential of AI to enhance the scalability in qualitative research while challenging its interpretive depth. It situates this tension within the historical trajectory of qualitative research -and specifically Grounded Theory- from positivist to constructivist paradigms, highlighting how AI's automated, data-driven approaches may signal a resurgence of positivist assumptions. Key research questions guide this exploration: To what extent do qualitative researchers harness AI's efficiencies in data analysis? Can the extended use of AI in qualitative research impact the depth and reflexivity essential to interpretive analysis? To delve into these questions the study employs a Technology Acceptance Model (TAM) survey combined with semi-structured interviews, strategically targeting European researchers to explore AI's perceived usefulness, ease of use, and implications for qualitative methodologies. Survey and interview findings reveal a generational divide: early-career researchers embrace AI's capacity for large-scale data analysis and thematic identification, while experienced researchers express scepticism about its impact on qualitative reflexivity and contextual richness. This generational gap implies that the receptiveness of younger researchers could lead to a gradual return to a methodological positivism. While this study brings a generational divide under the spotlight, future directions call for deeper investigations into the structural inequalities shaping AI adoption, such as access to resources, geography and gender.

Keywords

artificial intelligence, grounded theory, positivism, constructivism, qualitative research

Introduction

The integration of artificial intelligence (AI) in qualitative research marks a pivotal transformation in social research. Qualitative research, emerged primarily in the mid-20th century as an independent scientific field, emphasizing the need to understand social phenomena through rich, contextualized data rather than numerical abstraction. Initially rooted in anthropology and sociology, qualitative methods sought to capture the lived experiences, beliefs, and meanings individuals assign to their surroundings through reflective and participatory research processes (Denzin & Lincoln, 2011). Grounded theory (Glaser & Strauss, 1967), ethnography (Atkinson et al., 2001), and narrative analysis (Riessman, 1993) developed as core methods, providing frameworks for interpreting textual, visual, and auditory data.

Foundational principles in all these methodologies underscore interpretivism, subjectivity, and the belief that reality is socially constructed (Guba & Lincoln, 1994).

Despite its merits, traditional qualitative analysis is time-intensive and relies heavily on the researcher's interpretive skills. Researchers manually code transcripts, identify themes, and ensure credibility through constant reflexivity (Creswell & Poth, 2018). By implication, they face limitations when

¹School of Political Sciences, Aristotle University of Thessaloniki, Thessaloniki, Greece

Corresponding Author:

Georgios Chatzichristos, School of Political Sciences, Aristotle University of Thessaloniki, Leof 3 Septemvriou, Thessaloniki 54124, Greece.
Email: chatzichristos@polsci.auth.gr



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analysing large datasets. This is where AI's computational efficiency becomes relevant. AI's evolution, particularly over the last decade, has dramatically expanded its capabilities from basic data processing to advanced Machine Learning (ML) and Natural Language Processing (NLP) techniques. Pioneers like [Turing \(1950\)](#) speculated about machines' potential to simulate human thought, which laid the groundwork for AI's development. Advances in ML, coupled with increased computational power and the availability of big data, have enabled AI systems to "learn" from vast datasets and perform complex tasks such as image recognition, language translation, and predictive analytics ([Goodfellow et al., 2016](#)).

For qualitative research, AI tools can automate labour-intensive processes such as transcription, theme identification, and data analysis. As a result, AI tools are now highly relevant for qualitative research, addressing the need for speed and scalability while aiming to preserve interpretive depth. However, there are only a handful of studies that explore the importance of AI for qualitative research ([Marshall & Naff, 2024](#)), while its wider implications for qualitative theorising remain largely unexplored. The novel contribution of this study extends beyond the practical implications of AI adoption among qualitative researchers to a deeper exploration of its epistemological implications. The current study delves into this discussion by setting the following research questions. *To what extent do qualitative researchers harness AI's efficiencies in data analysis? How can the extended use of AI in qualitative research impact the depth and reflexivity essential to interpretive analysis?* To address the above questions, the study employed a Technology Acceptance Model (TAM), which was distributed among qualitative researchers across Europe to assess the acceptance and usage of AI in qualitative research. Existing studies on AI adoption in qualitative research remain scarce and limited in scope, with most focusing on descriptive analyses rather than in-depth examinations of its methodological and epistemological implications. Those available predominantly adopt a descriptive approach, emphasizing the disparities in AI adoption between different institutions ([Marshall & Naff, 2024](#)). The current study has a more in-depth perspective by dwelling on the balance between AI usage and interpretive reflexivity and this constitutes its novel contribution.

In order, to appreciate the significance of AI's role in this domain, it is crucial to explore the foundations of qualitative research methods, examine AI's technological evolution, review current scholarly discussions on AI's application in qualitative settings, and introduce novel research to inform this inquiry. Along these lines, the rest of the article is structured as follows. The following section reviews the literature on qualitative research tracing the path from positivism to constructivism. The second part of the section explores the incorporation of AI into qualitative research and the potential backlash to positivism. The next section describes the research design and methodology of the study, while the fourth section

interprets the results of the research. The final part of the article develops a broader discussion on the significance of the results, the limitations of the research and highlights future research directions.

Qualitative Research: From Positivism to Constructivism

Grounded Theory (GT), based on the emblematic work "*The Discovery of Grounded Theory: Strategies for Qualitative Research*" by [Glaser and Strauss \(1967\)](#) has been considered a milestone for qualitative research. Based on the positivism of the post-World War II period, the two scholars diagnosed that social research during that time was mainly based on the verification of pre-existing theories, rather than re-theorizing ([Timonen et al., 2018](#)). As they pointed out "*since verification has primacy on the current sociological scene, the desire to generate theory often becomes secondary, if not totally lost*" ([Glaser & Strauss, 1967](#), p. 2). Following their words, this created "*an embarrassing gap between theory and empirical research*" ([Glaser & Strauss, 1967](#), p. 2). The contribution of classical GT to the qualitative -and more broadly social-research was catalytic, at a time when extreme positivism was undermining qualitative research. The work of Glaser and Strauss was a 'groundbreaking statement' as it laid the foundations for a systematization of qualitative research. The positivism of the method showed that qualitative analysis could be methodical, rigorous and structured ([Charmaz, 2006](#), p. 5). In this context, the basic analytical procedures of classical GT were aimed at building a theory on the basis of what 'really exists' in the data ([Annells, 1996](#), p. 385). The objectivism and positivism of the method were ultimately aimed at an inductive theorizing.

Despite the important contribution of classical GT, Strauss himself moved away from Glaser's positivism and created the first major schism within GT. He criticized the extreme positivism of classical GT ([Strauss & Corbin, 1998](#)) and turned the research and analytical interest from the discovery of empirically grounded theories to their construction ([Charmaz, 2000](#)). This variant, captured in the relevant literature as Straussian GT, combined the inductive with the productive method ([Heath & Cowley, 2004](#)), thus challenging the doctrine of abstaining from the literature before research is initiated (advocated by classical GT), pointing out the difference between an 'open mind' versus an 'empty mind' ([Jones & Alony, 2011](#)). Although Straussian GT also failed to eliminate the objectivism of classical GT, the incorporation of pre-existing theories as well as the researcher's subjective interpretation played an important role in re-theorising and shaped a different GT framework. Thus, while Strauss and Corbin retained the rigorous and scholastic coding framework, this was now underpinned by a philosophy of realism and symbolic interaction ([Kenny & Fourie, 2015](#)).

The secession of the Straussian GT paved the way for the constructivist reformulation of the theory. The role of the researcher in the process of knowledge creation and study has been a key starting point for Charmaz (2006, 2014) and the constructivist GT she introduced (Timonen et al., 2018). Drawing from Strauss and Corbin (1998), Charmaz rejected the classical idea that theory exists independently and what remains for the researcher is to discover it through an analytical research process. Instead, she argued that “*neither data nor theories are discoveries... we construct our grounded theories through our past and present engagements and interactions with people, perspectives and research practices*” (Charmaz, 2006, p. 10). Along this line of reasoning, she argued that GT should be used as a flexible, heuristic strategy rather than a standardised procedure (Charmaz, 2000, p. 15). Based on her reasoning, the researcher cannot become a neutral observer but instead inevitably exerts an interpretive influence on the analysis of the data (Bryant, 2007). Constructivist GT has fully re-imagined the qualitative epistemology by advocating for an heuristic approach rather than a strict methodology. This constructivist shift according to Bryant (2007) saved “*the basic ideas of the method*”. Until recently, the trajectory of qualitative research appeared to shift from positivism toward constructivism. However, the integration of AI into the field seems to have altered this course. By emphasizing automated, repeatable, and data-driven approaches to theory formation, the use of AI in qualitative research may signal a potential resurgence of the positivist principles characteristic of classical GT.

AI in Qualitative Research: Back to the Positivism of Classical GT?

Scholarly interest in integrating AI into qualitative research has grown, especially in the last few years: most of the focus has been on how the use of AI improves data analysis (Hamilton et al., 2023; Morgan, 2023; Parker et al., 2023), facilitates virtual research (Stafford et al., 2024), as well as raises ethical and deontological considerations for social research (Chubb, 2023). At present, deep learning models (e.g., GPT) may not necessarily provide reliable information (Liu et al., 2023), thus leading qualitative researchers to erroneous propositions and false systematic reviews. In parallel, they might be even used to generate responses in interviews or online focus groups, thus misleading the researchers (Stafford et al., 2024).

Despite the challenges, AI-driven data analysis and coding have gained significant momentum in qualitative research. Researchers can now prepare data and conduct preliminary analysis more efficiently, thanks to AI-powered transcription tools like Otter.ai and Descript, which significantly reduce the time required for transcription. Simultaneously, machine learning (ML) and natural language processing (NLP) techniques—with ChatGPT being among the most

prominent—enable researchers to analyze vast volumes of qualitative data with increased regularity and accuracy. However, as AI development involves creating algorithms that mimic human cognitive abilities (Zhang et al., 2023), a critical question arises: To what extent can AI replicate human interpretative action, and what are the implications for qualitative analysis? This debate has generated diverse perspectives in the academic community.

On the one hand, studies suggest that large language models (LLMs) can aid qualitative researchers by deductively coding transcripts, offering a systematic and reliable platform for code identification, and mitigating potential misalignment in analysis (Tai et al., 2024). In this regard, AI can significantly enhance the efficiency of qualitative data analysis, particularly through automated transcription and coding processes (Saldaña & Omasta, 2018). Notably, ChatGPT has been shown to not only match human-coded themes in thematic analysis but also identify additional themes that researchers had initially overlooked (Parker et al., 2023). This has led to the recognition that AI tools like ChatGPT can serve as powerful complements to labor-intensive, human-centered research, ultimately streamlining qualitative workflows (Hamilton et al., 2023). On the other hand, concerns have been raised about the risks of inaccuracy, particularly when researchers lack familiarity with the data or are not proficient in prompt engineering (Chubb, 2023). Comparative studies between human and AI-generated thematic analyses reveal that while human coders identify certain themes that AI misses, the reverse is also true. A key distinction remains: ChatGPT excels at identifying concrete, descriptive themes but struggles with deeper, interpretative ones (Morgan, 2023). This underscores the necessity of human oversight in ensuring that AI-generated themes align with the contextual and nuanced nature of qualitative research. This way AI can enhance interpretation when used in tandem with human researchers, maintaining both efficiency and analytical rigor (Williamson et al., 2025). To navigate this process, recent methodological advancements propose hybrid approaches that integrate computational techniques with human expertise, such as Computational Grounded Theory (CGT) (Nelson, 2020).

However, the integration of AI into qualitative research also raises concerns about a potential return to positivist tendencies. The positivist underpinnings of traditional GT, which prioritize observable and repeatable patterns above subjective interpretations, are reflected in ChatGPT’s dependence on organized, data-driven methods. The objectivist and positivist goals of *discovering what really exists* in the data are echoed by these technologies, which can automatically identify descriptive concepts and themes in the literature. Furthermore, the positivist goals of classical grounded theory to forecast events based on data-driven insights are further aligned with AI’s prediction capabilities. This multifaceted alignment draws attention to a conflict in qualitative research: while modern approaches are moving more and more toward constructivist viewpoints that take complexity and context

Table 1. Research Design.

Research stage	Stage 1	Stage 2	Stage 3
Research Objective(s)	Exploring the acceptance of AI by qual researchers	In-depth investigation of the AI use by qual researchers	Data integration & interpretation
Research Tool	Technology Acceptance Model (TAM) survey	Semi-structured interviews	Thematic Analysis
Research subjects/ objects	Qualitative Researchers	Qualitative Researchers	

into account (Charmaz, 2021), AI tools like ChatGPT are fundamentally more positivist because of their reliance on quantifiable and repeatable results (Christou, 2023).

Therefore, incorporating AI into qualitative research can unintentionally return attention to positivism's methodical precision, possibly at the price of the subtle insights that interpretive themes offer. This brings up significant issues about how qualitative approaches might strike a balance between the depth and reflexivity essential to interpretive analysis and the efficiencies provided by AI. These worries also draw attention to the ethical difficulties of utilising AI as a research assistant in sensitive qualitative settings (Chubb, 2023), especially in light of the fact that AI systems have been demonstrated to reinforce prejudices such as racial and gender discrimination (Varsha, 2023). Accordingly, an AI-driven, renewed focus on positivist accuracy can impede the creation of innovative theoretical frameworks that might confront and dismantle ingrained preconceptions (Christou, 2023).

Research Design and Methodology

Data Generation

The research underpinning this study deployed a qualitative research design to better comprehend how researchers accept and integrate AI tools into qualitative research. It combined an adapted Technology Acceptance Model (TAM) survey with semi-structured interviews (see Table 1). The TAM is a theoretical framework that explains how users come to accept and use technology. It focuses on two primary factors: perceived ease of use and perceived usefulness, which influence users' attitudes and their intention to adopt technology (Davis et al., 2024). In contemporary settings, TAM is widely used to evaluate the acceptability of new technologies, including AI (Li, 2023), by understanding how these factors impact user behavior. The epistemology behind the TAM is largely positivist, emphasizing measurable, objective constructs.

The novel idea in this research design lies in the way it refines a positivist epistemology to guide a qualitative inquiry. By incorporating TAM into a qualitative framework, this study uniquely utilizes a positivist lens to structure data generation, essentially blending the objectivity typically associated with quantitative methods into the interpretive, subjective nature of qualitative research. This method serves as a methodological experiment in itself, testing how

effectively a positivist approach can be combined with qualitative reflexive research. Also, whether this hybridization can yield novel insights not only to AI adoption but qualitative theory development per se. In the first stage, a TAM model was modified to include variables specifically relevant to AI, such as perceived usefulness, perceived ease of use, trust in AI-generated insights, and perceived ethical concerns, measured on a Likert scale (1–10). These factors provided insights into how researchers view AI's role in their work. In the second stage, 15 semi-structured interviews were conducted to interpret and contextualize the survey results.

More analytically, the survey was developed using the EU Survey platform and distributed to qualitative researchers through a strategic sampling. This approach combined criterion sampling, targeting researchers actively engaged in qualitative research, with snowball sampling, leveraging multiple academic networks (Horizon Projects and COST actions) around the authors with qualitative researchers from various countries (Greece, Cyprus, Portugal, Romania, Germany, Spain, Italy, Denmark and Slovenia). The study acknowledges that respondents came from diverse disciplinary backgrounds, epistemological traditions, and methodological orientations. Rather than treating them as a homogeneous group, the research design took steps to ensure diversity in the sample, capturing a range of perspectives on AI adoption in qualitative research. However, the survey did not exclusively target researchers working with GT. While GT served as a key theoretical lens in interpreting AI's potential impact on qualitative epistemology, the participants included researchers employing various qualitative methodologies, such as ethnography, phenomenology, narrative analysis, and discourse analysis. This broader inclusion allowed the study to assess whether AI's integration influenced qualitative research epistemologies across different methodological traditions, rather than solely within the GT framework. The strategic selection of participants also considered researchers' fields of study, ensuring a mix of social sciences, humanities, and interdisciplinary research. While a more focused sample of only GT scholars might have strengthened the study's alignment with specific theoretical debates, the inclusion of a broader pool of qualitative researchers provided a richer comparative perspective.

The survey was distributed to an estimated sample of 500 qualitative researchers across Europe, yielding 93 responses between May and September 2024 after two rounds of

reminders, reflecting both a strong response rate and significant interest in the topic. While 93 responses may seem limited for a survey, it is important to recognize that simple descriptive statistics, in this context, align with qualitative research principles rather than aiming for broad generalizability (Chatzichristos & Hennebry, 2023). The objective of this study was not to extrapolate findings to the entire qualitative research community but to identify trends, correlations, and patterns in AI adoption within this targeted group. Given its exploratory nature, which prioritizes understanding researchers' attitudes over statistical representation, the sample size offers valuable insights into the perceptions and concerns surrounding AI integration in qualitative research.

The subsequent interviews further enriched the findings by providing a deeper exploration of the themes that emerged from the survey. A total of 15 semi-structured interviews were conducted to interpret and contextualize the survey results. While this number falls on the lower end of Bryman's (2016) suggested range of 15–30 interviews in qualitative research, it was guided by the principle of theoretical saturation, ensuring that data collection continued until no significant new insights emerged. This approach aligns with qualitative research best practices, where depth and richness of data take precedence over sheer sample size. A strategic sampling strategy was used to select key informants (researchers that had already filled in the survey), based on diversity in years of experience, openness to AI, and demographics. This sampling strategy ensured that the research findings were meaningful in the specific context of qualitative AI adoption, balancing breadth (survey) and depth (interviews) to address the study's objectives. The interview protocol was designed to explore the fundamental epistemological concerns raised by AI adoption in qualitative research. Specifically, it investigated how AI's efficiency and scalability might conflict with the depth, reflexivity, and interpretive richness that define qualitative methodologies. The interview questions were structured around key themes emerging from the survey, including perceptions of AI's role in theory development, concerns about its influence on reflexivity, and attitudes toward its potential shift toward positivist paradigms. Interviews, conducted via video conferencing between September 2024 and November 2024, were recorded, transcribed verbatim, and anonymized for ethical compliance.

Data Analysis

The analysis of the survey involved a multi-step approach to examine correlations and trends within survey data related to the use of AI in qualitative research as well as demographic variables (e.g., years of qualitative research experience, age, and gender). A Pearson correlation matrix was computed to identify relationships between variables¹ (see Appendix Figure 5). To focus on meaningful correlations, redundant self-correlations and duplicates were removed, and significant correlations were highlighted for further analysis. For each

significant correlation, simple linear regression was conducted using the Ordinary Least Squares (OLS) method to predict the dependent variable based on the independent variable. This methodological approach combined descriptive statistics, correlation analysis, and regression modeling to trace patterns, trends, and relationships within the survey results (see Appendix Table 2). Visualizations like scatter plots and heatmaps were employed to support the statistical findings and enhance interpretability.

While regression analysis was employed to explore correlations between various survey variables, this study cannot be classified as mixed-methods research. Qualitative methods supported by simple descriptive statistics do not warrant the mixed-methods label, as this classification is commonly misused in such contexts. Instead, such approaches should be appropriately identified as qualitative research, aligning with the argument that qualitative studies supplemented by basic statistical insights maintain their qualitative essence rather than transitioning into mixed-methods frameworks (Chatzichristos & Hennebry, 2023). A thematic analysis approach was used to identify patterns and themes in the semi-structure interviews. Using the six-phase process outlined by Braun and Clarke (2006), we began by thoroughly familiarizing ourselves with the interview transcripts through multiple readings. We then generated initial codes by identifying recurring patterns and salient points related to AI adoption, interpretive challenges, and methodological concerns. These initial codes were systematically collated into candidate themes, which were then reviewed and refined through an iterative process involving independent coding by multiple team members. Ultimately, key themes such as "Efficiency and Scalability of AI Tools," "Ease of Integration and User-Friendliness," "Ethical Concerns and Trust," and "Impact on Reflexivity and Positivist Bias" emerged as central to the discourse. These themes not only structure our research findings but also provide a critical lens through which we examine whether the systematic, data-driven approach of AI may inadvertently steer qualitative research toward the objectivity and repeatability characteristic of classical Grounded Theory.

Key findings were triangulated by integrating survey and interview data to validate statistical correlations and provide rich contextual insights. For instance, our regression analysis indicated that experienced researchers were more skeptical of AI's capacity to generalize theories, a trend that was further illuminated during interviews. Participants' narratives and direct quotes helped explain this skepticism by revealing concerns about reduced interpretive depth and the risk of adopting a positivist bias—hallmarks of classical Grounded Theory. By linking these qualitative themes, such as "Impact on Reflexivity and Positivist Bias," with quantitative trends from the survey, our methodological design enabled a robust, multi-layered interpretation of AI's role in qualitative research. This integrated approach not only confirms the statistical findings but also uncovers deeper insights into researchers' experiences and attitudes, demonstrating how the

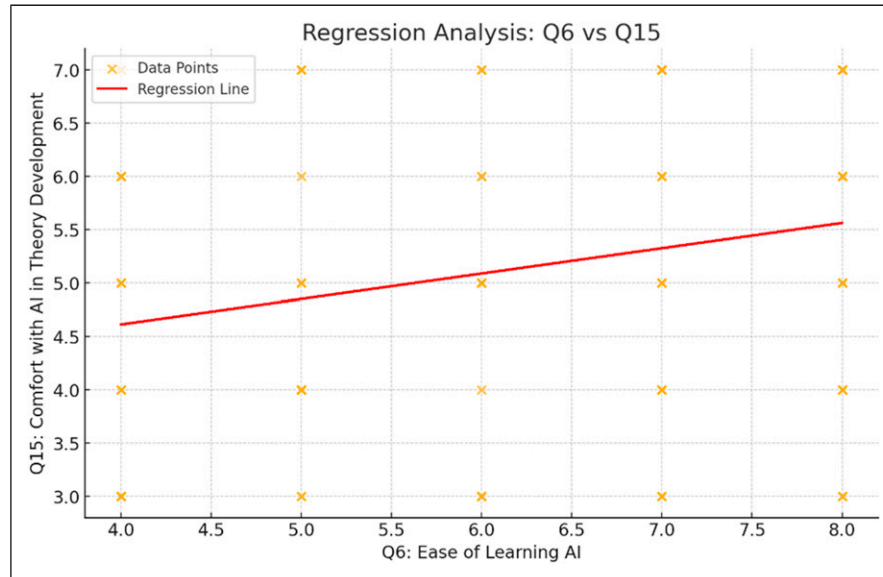


Figure 1. Regression Analysis Between Q6 and Q15.

systematic, data-driven nature of AI might subtly shift qualitative inquiry toward a more objective, repeatable paradigm.

Acceptance of AI in Qualitative Research

The research findings provide a nuanced understanding of researchers' perceptions of AI in qualitative research, combining quantitative survey results with insights from semi-structured interviews. Key themes emerged across Perceived Usefulness (PU), Perceived Ease of Use (PEOU), Attitudes Toward AI (ATU), and Behavioral Intention to Use (BIU), highlighting both opportunities and concerns regarding AI integration. Survey results indicated that many researchers value AI for its ability to handle large datasets and identify patterns and themes and this points to the potentiality of AI to enhance qualitative research: those who see AI as helpful for managing large datasets (Q4) also tend to believe that AI can generate more predictive and generalizable theories (Q5) (0,27 correlation). This connection highlights a perception that AI's capacity for handling large-scale data can translate into broader research insights and findings, underscored by a strong consensus among survey respondents on the utility of AI in qualitative research. Interviews reinforced this view, as participants noted AI's utility in overcoming human limitations: "AI can process thousands of pages of text in minutes, something I couldn't even imagine doing manually" (Participant 5, 8 years of experience).

Ease of use emerged as another significant factor influencing comfort with AI's role. Respondents who find AI tools easy to learn and integrate (Q6) are more comfortable with the idea of AI playing a central role in theory development (Q15) (0,24 correlation, see Figure 1). This suggests that ease of use

within the TAM framework is a significant factor influencing positive attitudes toward AI's role in research. Interviewees emphasized this learning curve: "Once I got the hang of it, AI tools became part of my routine. But getting there wasn't easy" (Participant 8, 3 years of experience). Others, however, described challenges with tool interfaces and technical support: "Most AI tools are built for engineers, not social scientists. I struggled to make them fit my workflow" (Participant 3, 10 years of experience). These references seem to echo a need for trainings on AI.

Despite reservations, behavioural intention to use AI tools was generally positive and most researchers were quite receptive to participating in AI trainings (8,57). Along this line, participants open to training highlighted the need for ongoing skill development and not merely tutorials: "To use AI effectively, we need tailored training for researchers, not generic tutorials" (Participant 11, 7 years of experience). Contrary to intuitive assumptions, a negative correlation between Q2 (AI's pattern identification) and Q19 (openness to training) revealed that those confident in AI's capabilities were less likely to pursue additional training (see Figure 2). Interviewees confirmed this by stressing that "If AI does what I need, why would I sit through hours of training?" (Participant 6, 5 years of experience). In other words, AI was approached as a tool to reduce workload, and training was often perceived as an additional part to this workload. If AI was perceived as generating more training needs, then it was approached with higher skepticism.

While many saw AI as a powerful asset, there was a notable worry about its impact on interpretive approaches, bringing the shift towards the positivistic paradigm under the spotlight. Notably, one participant highlighted that "AI feels like it pushes us toward a more positivist mindset, and that's not

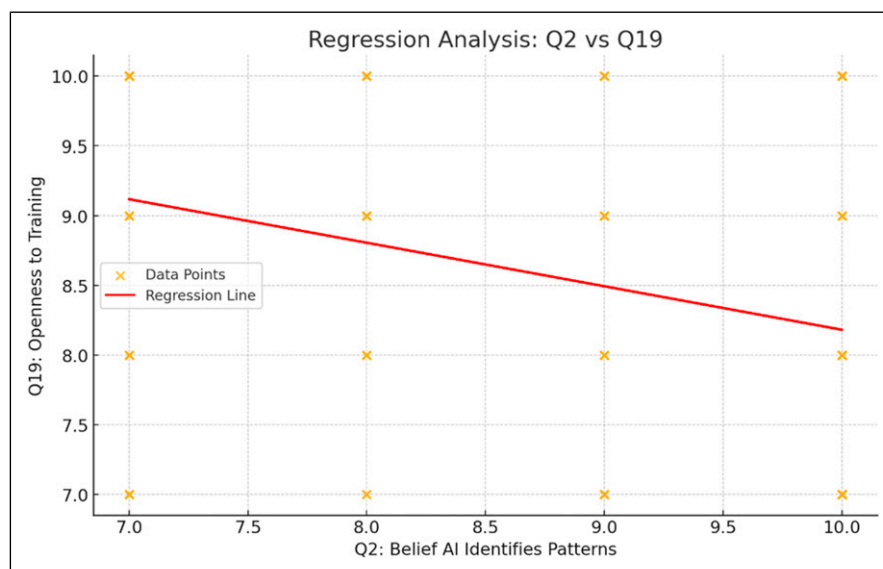


Figure 2. Regression Analysis Between Q2 and Q19.

what qualitative research is about” (Participant 9, 12 years of experience). In the same vein, concerns about AI shifting focus from interpretation to validation were particularly strong among experienced researchers. As one eloquently described *“I fear we’re replacing human judgment with data outputs that we don’t fully understand”* (Participant 14, 17 years of experience). Empirical findings from the interviews revealed a clear divergence in how researchers viewed AI’s epistemological impact. Early-career researchers, who were generally more open to AI, tended to frame it as a tool for increasing efficiency, particularly in handling large datasets and identifying patterns. However, experienced researchers expressed concern that AI’s emphasis on structured, repeatable, and data-driven processes could inadvertently reinforce positivist assumptions, limiting the interpretive flexibility necessary for qualitative inquiry. A senior researcher explained, *“When we engage with our data, we don’t just extract themes—we negotiate meaning through reflexivity and iterative analysis. AI lacks this kind of relational engagement; it sees words, not experiences”* (Participant 7, 18 years of experience).

The findings also highlighted tensions between AI’s role in coding and thematic analysis. While some researchers saw AI-assisted coding as a valuable complement to human-led analysis, others feared that it might standardize and constrain the interpretive process, diminishing the role of human reflexivity. As one mid-career researcher remarked, *“AI-generated codes might be accurate, but they lack nuance. It’s one thing to categorize a sentiment as ‘positive’ or ‘negative,’ but what about irony, implicit biases, or cultural subtext? AI doesn’t engage with these complexities”* (Participant 11, 7 years of experience). This skepticism was particularly evident in concerns about AI’s inability to engage with the nuances of lived experiences, power dynamics, and cultural contexts—core elements of many qualitative

methodologies. One interviewee, who specialized in ethnographic research, expressed frustration with AI’s limitations: *“In my fieldwork, context is everything. The way someone pauses before answering, the subtle shifts in tone—these details shape interpretation. AI can’t capture those human subtleties, which makes me question how much we should really trust its coding”* (Participant 14, 17 years of experience). Ultimately, these findings suggest that AI adoption in qualitative research is not simply a methodological shift but an epistemological one, with implications for how knowledge is constructed and validated.

This aligns with the survey results which indicate that more experienced qualitative researchers (years) are less likely to believe that AI leads to predictive and generalizable theories (Q5). More analytically, for every additional year of experience, the Q5 score decreased by approximately 0.06 units and this relationship proved to be statistically significant (see Figure 3). This could reflect skepticism among experienced researchers who value the interpretive and contextual nature of qualitative research over AI-driven generalizations, suggesting that experienced researchers are less convinced of AI’s ability to generate meaningful, context-sensitive theories. This was also echoed by the interviews with a more experienced researcher stressing that *“the richness of qualitative data lies in interpretation, not prediction. AI might lose that nuance”* (Participant 12, 15 years of experience). Another quote that echoes this sentiment came from a mid-career researcher who expressed similar concerns: *“AI can process data faster than any human, but it doesn’t understand the context or the story behind the numbers. That’s where we need qualitative research”* (Participant 7, 18 years of experience).

The negative correlation (-0.215) between years of qualitative research experience and the belief that AI tools can enhance the accuracy of qualitative data analysis by

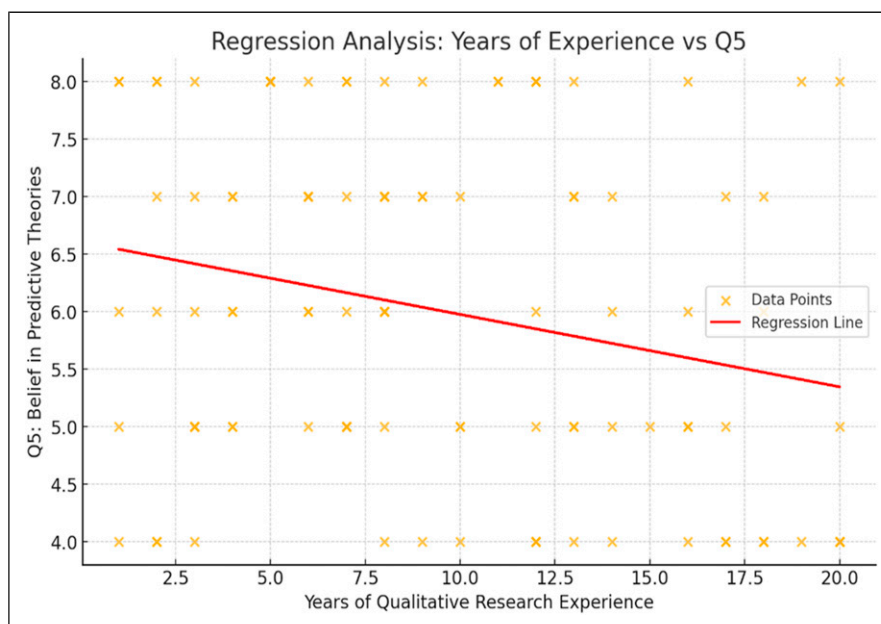


Figure 3. Regression Analysis Between Years and Q5.

minimizing subjective biases (Q1) underscores once more the importance of the experience of qualitative researchers (see Figure 4). This skepticism seems to stem from a deeper familiarity with the complexities of qualitative data and the interpretive nuances that AI tools may overlook. Experienced researchers seem to have a stronger feeling on the inherent and valuable aspect of subjective bias in the qualitative inquiry. Driven by human judgment and contextual understanding this subjective bias cannot be fully replicated by AI. As one interviewee noted, *“AI can standardize processes, but it doesn’t eliminate the biases embedded in the data or the framing of questions”* (Participant 9, 12 years of experience). This finding highlights once more a tension where AI’s perceived objectivity is not universally accepted, particularly by more experienced researchers who prioritize the richness of human interpretation in qualitative methodologies.

Eventually, younger researchers and those with less experience tend to view AI as a valuable tool for streamlining workflows, often emphasizing its utility in reducing workload and handling complexity. However, experienced researchers, who seem to value more the interpretive and contextual richness of qualitative data, express apprehension about AI’s alignment with qualitative epistemologies, viewing it as potentially steering research toward a positivist paradigm.

Discussion

In seeking to illustrate the extent to which qualitative researchers are leveraging AI in data analysis, this study showed that while AI is widely recognized for its ability to enhance efficiency, manage large datasets, and identify patterns that might otherwise be overlooked, its integration into qualitative

research presents a complex interplay between opportunity and skepticism. This interplay manifests itself in a generational divide, with early-career researchers being more comfortable with its role in theory development, but experienced researchers remain skeptical. Their concerns relate to AI’s ability to shift the focus from reflective, human-centric interpretation to algorithm-based validation. This skepticism was reflected in both survey data -such as the negative correlation between years of experience and belief in AI’s predictive capabilities- and interviews, where participants voiced fears about replacing human judgment with outputs lacking contextual depth. Early-career researchers, who are less skeptical and more open to AI tools, show a greater willingness to adopt and trust AI-driven processes. This generational divide suggests that the tension between AI and qualitative research traditions could intensify over time, as younger researchers’ openness may materialize a gradual shift toward positivist approaches, potentially altering the epistemological core of qualitative inquiry. By implication, AI might indeed push qualitative inquiry toward a more positivist mindset and this would raise and ontological challenges for qualitative research.

Qualitative research holds a unique and irreplaceable role in efforts to raise social value (Chatzichristos & Perimenis, 2022) while at the same time promoting social justice (Reid, 2004). In parallel it aims at transforming research per se by decolonialising learning (Thambinathan & Kinsella, 2021). Due to that, it requires the active engagement and empathy of human researchers working collaboratively with communities (Chatzichristos et al., 2021), unlike AI-driven analysis, which lacks deep contextual insights. The role of qualitative researcher is not limited to that of a passive observer but extends

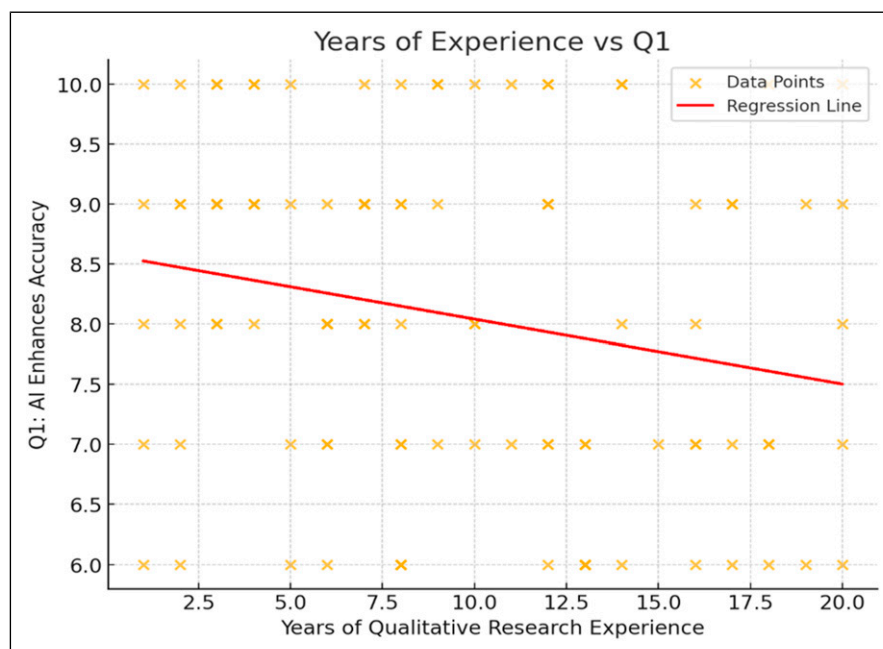


Figure 4. Regression Analysis Between Years and Q1.

to co-participating and being involved in iterative cycles of planning, acting, observing, and reflecting alongside community members. This process demands sensitivity to cultural nuances, evolving social dynamics, and complex human motivations—qualities that AI cannot fully replicate. The transformative potential of action research lies in its ability to foster trust, empower voices, and address power imbalances, facilitating real change within societies.

The dialogue seems to echo the internal GT considerations between the unbiased nature of classical GT and the constructivist perspective of GT: AI -similar to positivist assumptions- may be a valuable tool in qualitative research, but it cannot substitute the relational and participatory depth -the constructivist counterpart- that defines qualitative research as a catalyst for social change. This seems to be the fundamental distinction that the current study brings under the spotlight. The methodology underpinning this study demonstrates that a positivist epistemology can effectively complement reflexive and relational qualitative research, much like TAM supplemented the qualitative interviews. However, without the depth provided by in-depth interviews, TAM would merely indicate trends that could be misleading. For example, an increase in AI adoption might be superficially attributed to the researchers' age, without uncovering that age, in itself, is not a decisive factor. This research design serves as an experiment in its own right, assessing the extent to which a positivist approach can be integrated with qualitative reflexive research. Ultimately, this integration has the potential to establish a new paradigm for qualitative data analysis in the era of AI. Aligning with more practice-oriented studies this paradigm might include human, reflexive coding alongside AI deductive coding (Tai et al.,

2024), where human researchers and AI systems work in tandem to enhance interpretative depth while maintaining analytical rigor (Williamson et al., 2025). Still, even reflexive research cannot illuminate all aspects of a phenomenon. While it provides depth and context, certain underlying patterns or structural influences may remain obscured. Against this background, what was identified as a generational divide in the current study might, in essence, be a divide of resources. Early-career researchers often grapple with limited access to funding, constrained time due to precarious academic positions or the pressures of securing tenure, and fewer opportunities to build extensive research teams. These limitations can restrict their ability to engage in the kind of thorough, reflective qualitative research that demands substantial time and effort. On the other hand, well-established qualitative researchers—who have typically navigated the hurdles of early-career challenges—might enjoy greater access to institutional support, more robust funding opportunities, and established research teams. These advantages not only enable them to undertake reflective and nuanced research but also position them to set the standards and benchmarks for qualitative inquiry. Geography may also play a significant role in shaping access to resources and opportunities for researchers. Researchers in developing and underdeveloped countries have severe resource constraints (Heng et al., 2022) that limit their ability to meet research standards that are set -advertently or not- by European scholars. This might be grafted into a broader argument that stresses how new technologies might deepen economic inequalities and increase job insecurity (Avagianou et al., 2024). Gender further complicates this equation: persistent gender discrimination in academia frequently translates into

reduced access to funding, fewer leadership opportunities, and less institutional support for non-male researchers (Blithe & Elliott, 2019).

Against these backdrops, the identified divide may not be generational but systemic, rooted in broader structural inequalities that permeate academic institutions globally. What appears as a critique of the so-called “positivist turn” in qualitative research might instead reflect the elitist and male-dominated perspectives of those critiquing it – the authors being among them. This “male gaze” often dismisses the realities faced by under-resourced researchers, perpetuating an exclusionary framework that privileges the viewpoints of those who already benefit from academic privilege. In simple words, AI’s positivist turn might come along with a fairer resource allocation in qualitative research. These appear to be critical parameters that still remain outside the limited scope of this research. These are the basic limitations of the study, as the research was conducted in European developed countries, which may also explain why gender differences were insignificant. Nevertheless, they point to future research to be done, linking the acceptance of AI in qualitative research to all these structural parameters.

Conclusion

Commencing with an inquiry about the extent of AI usage in qualitative data analysis and how this usage can impact the depth and reflexivity of interpretive analysis, this study showed that while AI-driven tools enhance scalability and reduce workload, their reliance on automated, repeatable processes aligns with positivist paradigms, potentially reshaping the epistemological foundations of qualitative research. The findings suggested a clear generational divide: early-career researchers are more receptive to AI’s efficiencies, whereas experienced researchers remain skeptical of its potential to replace human interpretation. However, this divide may not be purely generational but rather reflective of broader systemic inequalities, including disparities in funding, geographic access to resources, and gender-based barriers in academia. The study underscored the need for critical engagement with AI in qualitative research, ensuring that technological advancements do not compromise the reflexivity, contextual richness, and participatory dimensions that define qualitative inquiry.

The implications of these findings are twofold. First, the growing integration of AI into qualitative research demands methodological innovation that bridges positivist epistemology with reflexive analysis while balancing AI’s computational efficiency with the nuanced interpretive agency of human researchers. The current research design might be a new paradigm in itself by utilizing a positivist epistemology to guide the qualitative data generation. Second, structural inequalities in academia must be addressed to ensure that AI’s adoption does not reinforce existing disparities but instead democratizes access to research tools. Future studies should explore how AI adoption intersects with issues of privilege, resource allocation, and academic hierarchies, shaping not

only the methodology of qualitative research but also the power dynamics within the research community itself. Ultimately, the recognition of these structural divisions is critical to fostering a more inclusive and equitable academic landscape, where the interaction between AI and qualitative research can take completely different directions. And one thing is for sure: the exact direction of this interplay could only be decided by non-artificial intelligence.

ORCID iD

Georgios Chatzichristos  <https://orcid.org/0000-0002-5004-5974>

Ethical Considerations

The AUTH Ethics Review Committee at AUTH University approved our interviews on Month 11, 2024. Respondents gave written consent for review and signature before starting interviews.

Consent to Participate

The study was approved by the AUTH Ethics Review Committee at AUTH University approved our interviews on Month 11, 2024. All participants provided written informed consent prior to participating.

Consent for Publication

Informed consent for publication was provided by the participant(s).

Author Contributions

Georgios Chatzichristos: Conceptualization; Methodology; Formal Analysis; Investigation; Writing – Original Draft Preparation; Writing – Review & Editing.

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Data is available by the authors upon reasonable request.

Supplemental Material

Supplemental material for this article is available online.

Note

1. Pearson’s correlation coefficient measures the linear association between two continuous variables, ranging from -1 (perfect negative correlation) to $+1$ (perfect positive correlation).

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